Dita Aulia - DABI 22

Take Home Test

CUSTOMERS CHURN **PROJECT**

from customers behavior to retention strategies



Hello, I’m DITA !

I enjoy exploring data, identifying valuable trends, and transforming analytical findings into actionable business insights.

Skills

- Analytical Thinking
- Problem Solving
- Communication
- Attention to Detail

Language

Bahasa Indonesia - Native
English - Advanced
Chinese - Elementary
Japanese - Elementary

Tools

Ms Excel

Postgree SQL

Google Colab

Power BI

Ms PowerPointl



Dita Aulia A

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EDUCATIONAL
background

Sepuluh Nopember Institute of Technology
Physics | Aug 2012 - Feb 2017
GPA : 3.05 / 4.00
Instrumentation - Building Physics -
Computation - Simulation

Dibimbing.id
March 2026 - present
Data Analyst and Business Intelligence
Excel - Python - SQL - Power BI
15+ Hands-On Projects

EXPERIENCE

Physics Teacher
Trimurti Senior High School Surabaya
July 2022 - present

International Partnership Coordinator | 2025

Finance Admin Support | 2023-2024
focus on schools' store and official merch
recorded and monitored daily income and expense
transaction. collected and evaluated customers'
needs and purchasing trends

PROJECTS ALONG THE WAY

STATIONERY STORE SALES ANALYSIS

Analyzed sales performance to identify top products, city-level
trends, payment preferences, and order status.
Tools: Ms Excel
Focus: Sales Analysis · Dashboard · Business Insights

Project's Link

CUSTOMER SEGMENTATION

Segmented customers using RFM Analysis to identify loyalty levels, customer value, and churn risk for targeted marketing strategies.
Tools: Power BI · DAX
Focus: RFM Analysis · Customer Segmentation · Business Insights

Project's Link

ETL & DATAMART IMPLEMENTATION

Built an ETL workflow and stored procedures to transform raw e-commerce data into structured datamarts for efficient analysis and reporting.
Tools: dBeaver (PostgreeSQL)
Focus: ETL · Stored Procedure · Datamart · Partitioning

Project's Link

SOCMED ENGAGEMENT ANALYSIS

Analyzed cross-platform social media performance to measure engagement, compare content effectiveness, and identify high-performing content patterns.
Tools: Google Analytic
Focus: Engagement Metrics

Project's Link



TODAY'S MAIN PROJECT

- Understanding why customers leave
- Customers churn analysis and retention insights

Project's Link



BUSINESS PROBLEM

Why are Customers Leaving?

Customer churn can affect revenue, customer relationships, and long-term growth. Identifying which customers are likely to leave and the factors linked to churn helps businesses develop more effective customer retention strategies.

Analysis Objectives

- Understanding Customer Churn
- Identifying Churn Patterns
- Determining Key Factors
- Driving Retention Actions

PROJECT WORKFLOW



01 Data Understanding

Review the available tables, examine data dimensions and column types, and assess missing values and duplicate records.

02 Data Prepocessing

Handle missing values, combine relevant tables, review duplicate columns, and prepare the dataset for analysis.

03 Exploratory Data Analysis

CConduct univariate and bivariate analysis, assess outliers, compare churn rates across customer groups, and examine numerical correlations.

04 Feature Engineering & Pattern Analysis

Create tenure groups and examine combinations of characteristics, such as contract type and tenure, to identify more specific churn patterns.

05 Insights & Recommendations

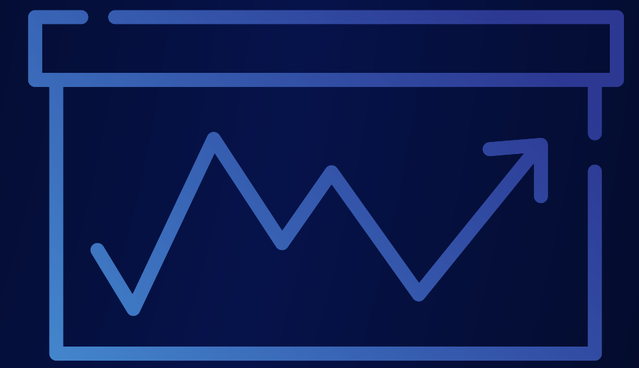
Summarize the strongest observed patterns and translate them into targeted customer retention recommendations.

UNDERSTANDING THE DATA

The dataset contains information that help explain customer behavior and identify patterns related to churn.

KEY VARIABLES

- DEMOGRAPHIC
  Age, Gender, Location, Senior Citizen Status
- CUSTOMER RELATIONSHIP
  Tenure, Contract Type, Customer Status
- SERVICE USAGE
  Phone Service, Internet Service, Additional Services
- CUSTOMER EXPERIENCE
  Satisfaction Score, Customer Lifetime Value (CLV)
- PAYMENT BEHAVIOR
  Monthly Charges, Total Revenue, Payment Method





CHURN » YES / NO



DATA PREPARATION

HANDLING MISSSING VALUE

Detected missing values in 3 datasets:

- Online Services: 1,526 missing values
- Service Options: 3,877 missing values
- Status Analysis: 5,174 missing values

H**andling:**

- Internet Type → filled with "No Internet Service"
- Offer → filled with "No Offer"
- Churn Reason → filled with "Not Churned"

DATA INTEGRATION

Combined 6 datasets

Validated customer uniqueness after merging:

- Total Customers: 7,043
- Duplicate Customer ID: 0

FEATURE PREPARATION

tenure Group

- 0–12 Months
- 13–24 Months
- 25–48 Months
- 49+ Months

Purpose:

To analyze churn behavior based on different customer tenure stages

FINAL CHECK

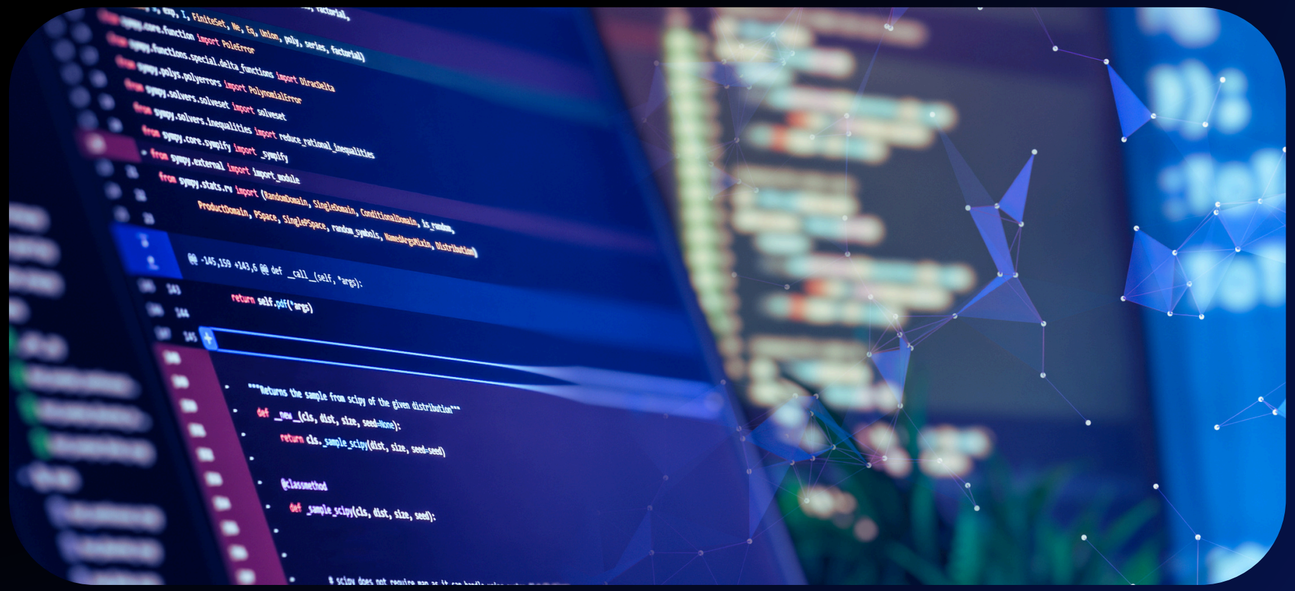
Rows: 7,043 customers

Features: 52 variables

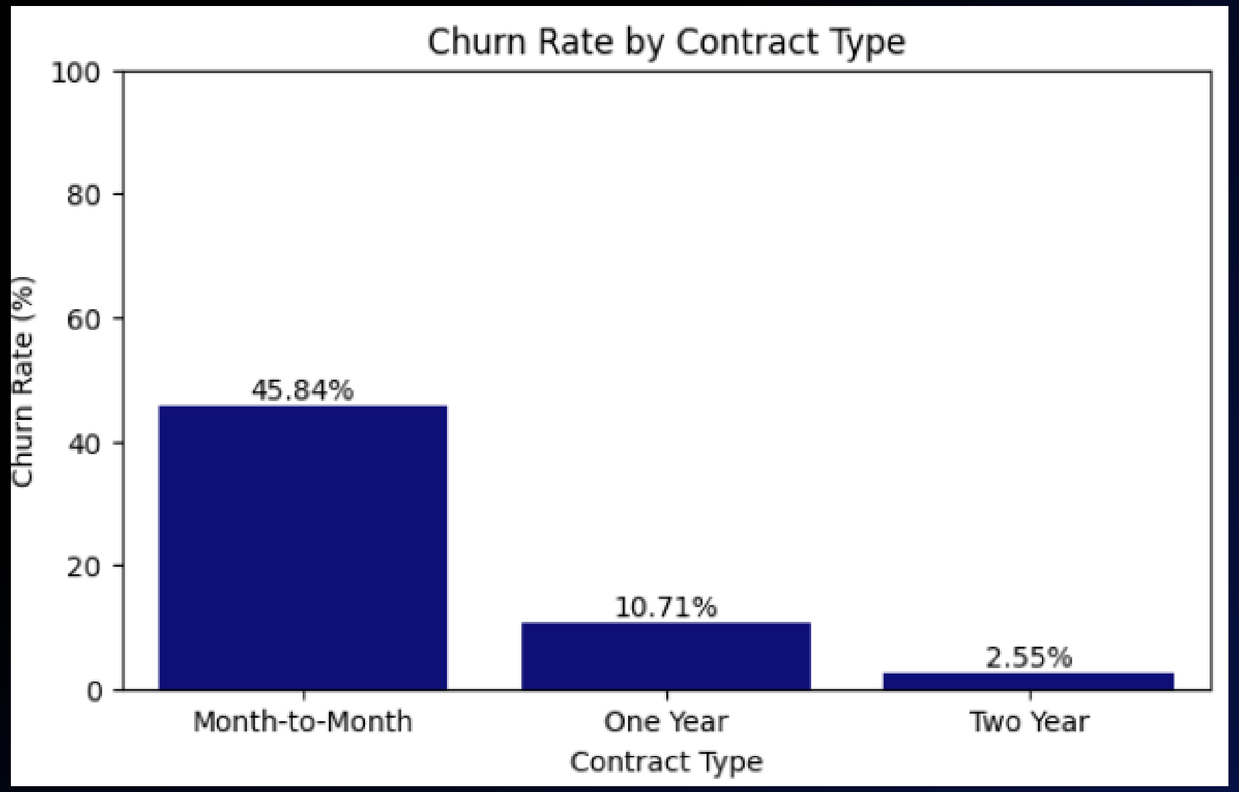
Missing Values: 0

Duplicate Customer ID: 0

UNDERSTANDING CUSTOMERS CHURN



Month-to-Month Customers Show Higher Churn Risk



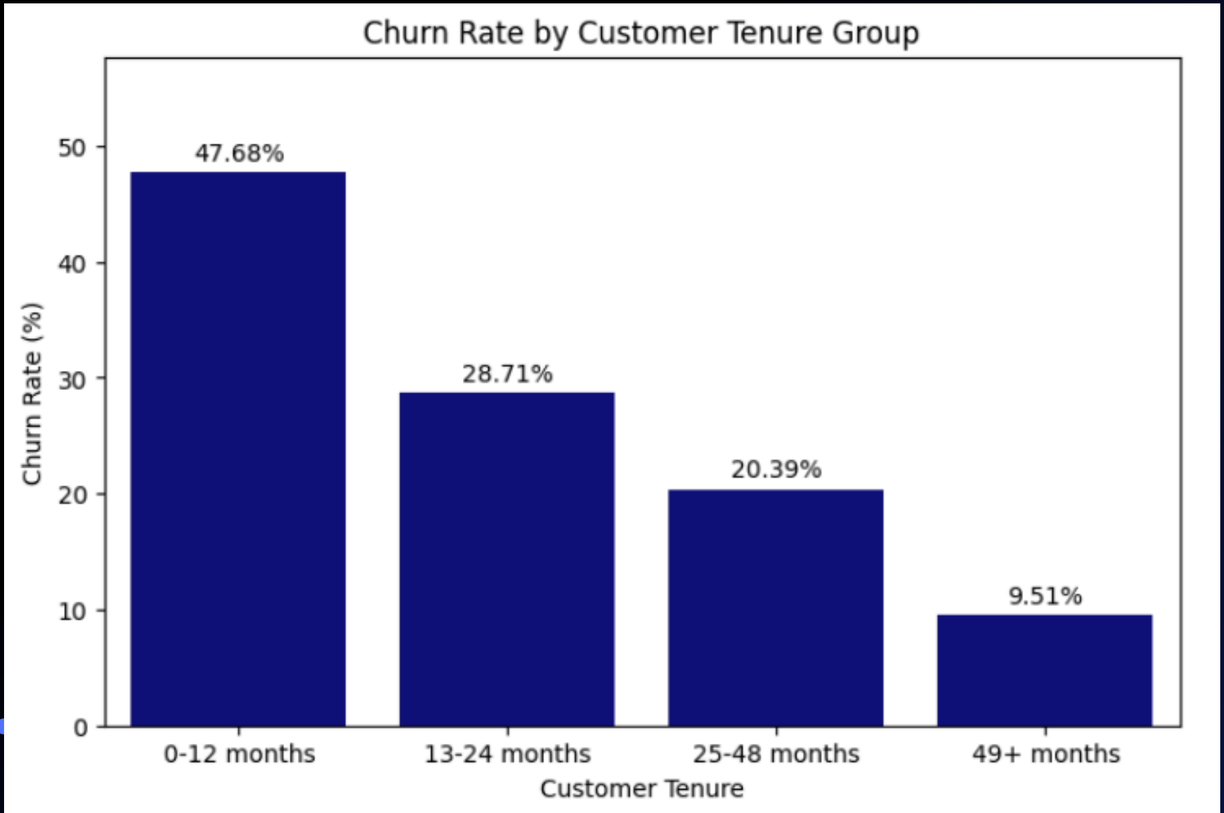
INSIGHT

Month-to-Month customers have the highest churn rate at **45.84%,** while Two-Year contract customers show the lowest churn rate at 2.55%.

INTERPRETATION

Customers with month-to-month contracts are more likely to churn than those with longer-term commitments. This suggests that targeted engagement and retention initiatives could help encourage these customers to maintain their subscriptions for longer periods.

The First Year Can Make or Break Customer Loyalty



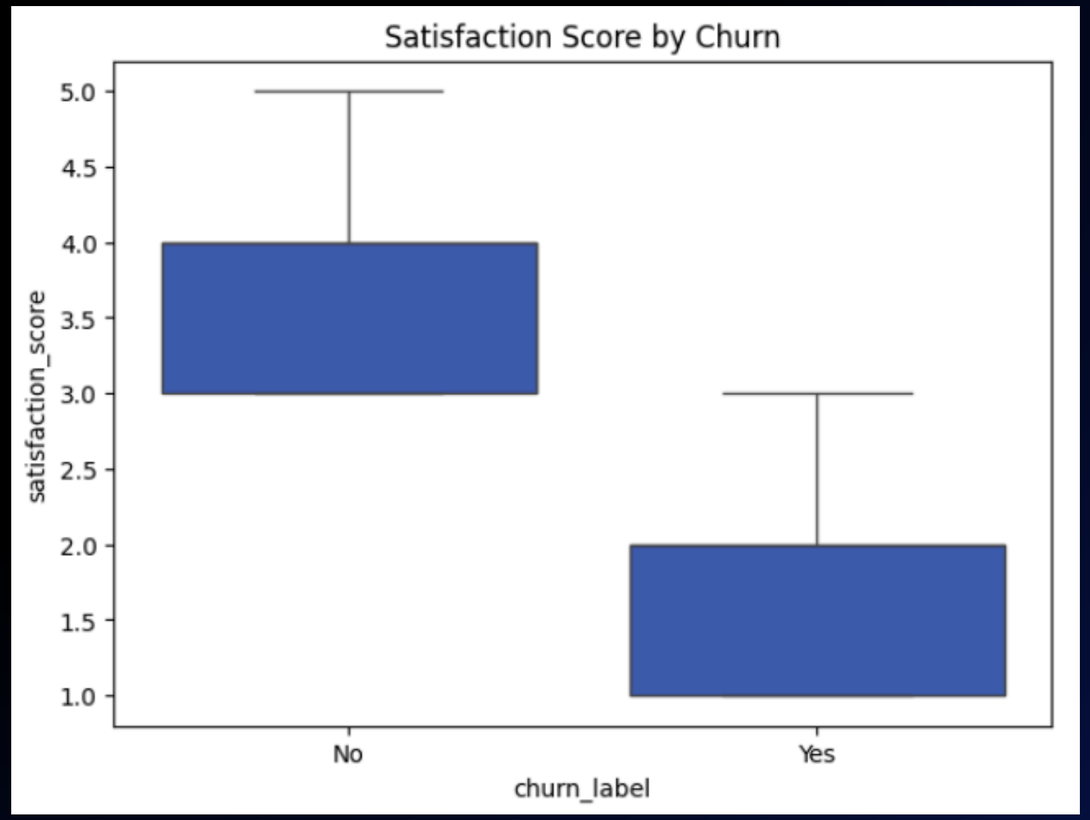
INSIGHT

Churn is highest among customers steadily declines as tenure increases, reaching just 9.51% among customers who have stayed for 49 months or more.

INTERPRETATION

The first year is a critical window for customer retention. Strengthening onboarding, improving the customer experience, and maintaining proactive engagement during this period may help reduce early churn and build long-term loyalty.

When Satisfaction Drops, Churn Risk Rises



INSIGHT

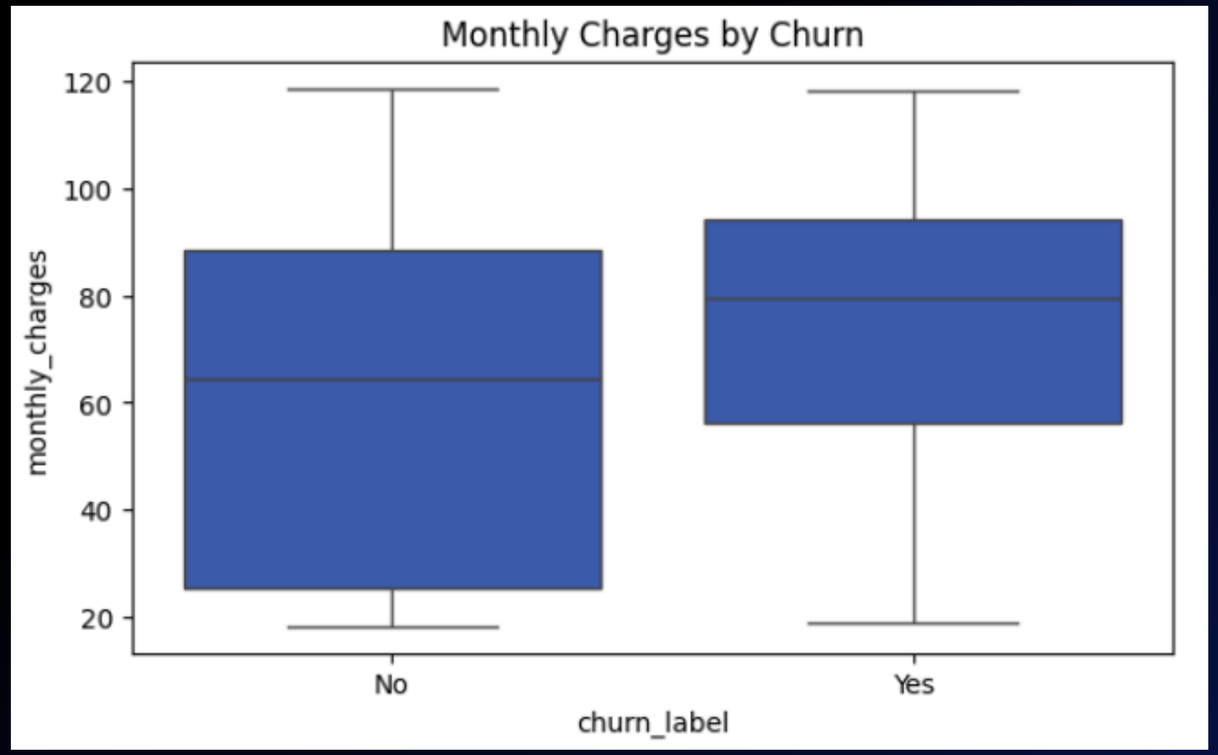
Customers with lower satisfaction scores show a higher tendency to churn.
Average Satisfaction Score;

- Churned Customers: 1.74
- Retained Customers: 3.77

INTERPRETATION

Customers reporting lower satisfaction tend to be at greater risk of churn, highlighting the need for targeted retention efforts to improve their experience and strengthen loyalty.

Higher Monthly Bills May Come with a Higher Risk of Churn



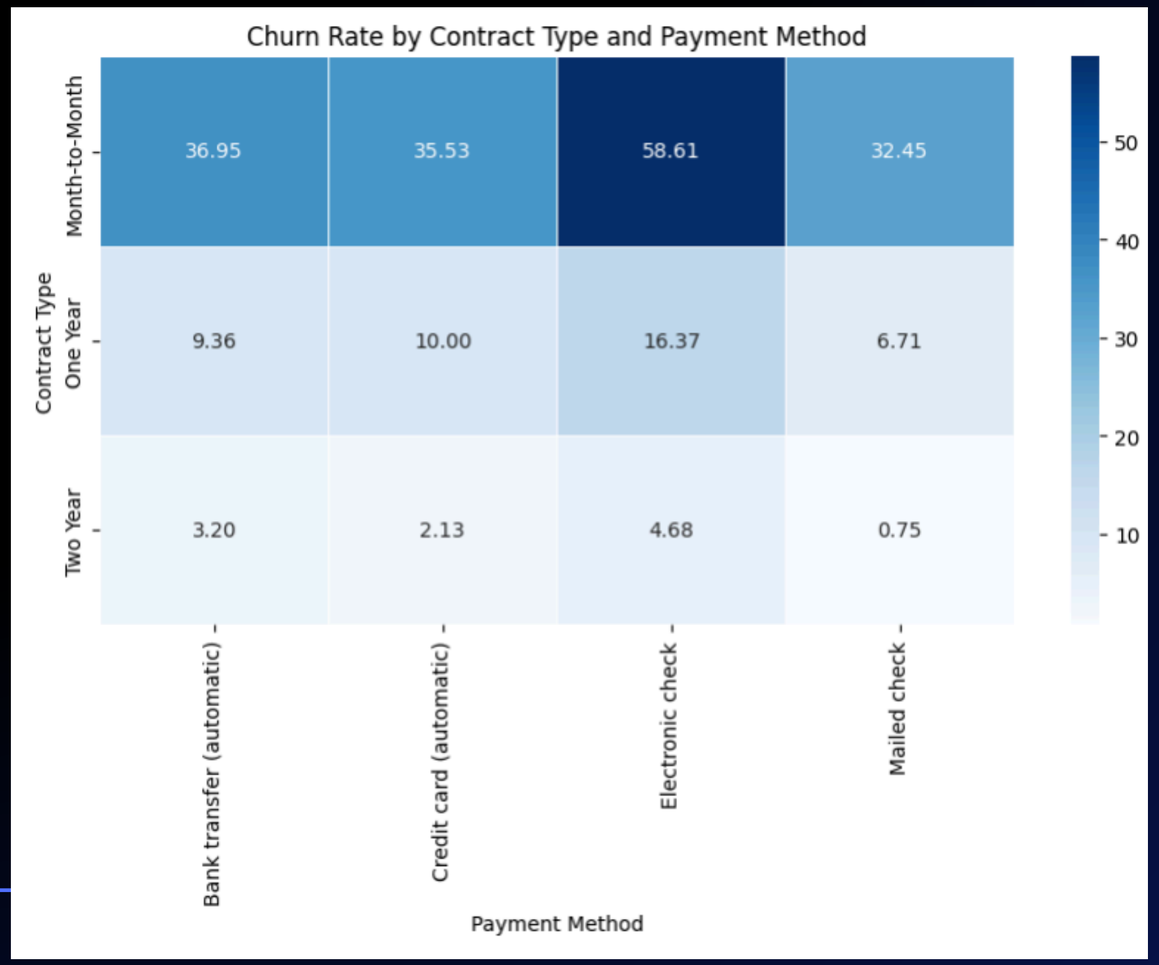
INSIGHT

Customers who churned had higher average monthly charges than those who remained with the company, with average charges of 74.44 for churned customers compared to 61.27 for retained customers.

INTERPRETATION

Higher monthly charges may reflect concerns about pricing, perceived value, or whether services meet customer expectations. Reviewing pricing plans and ensuring customers receive value that aligns with their spending may help strengthen retention.

The Combination of Contract Type and Payment Method Reveals a High-Risk Customer Segment



INSIGHT

Customers with both month-to-month contracts and Electronic Check payments exhibit a particularly high churn rate of approximately 58.61%, making this combination a key segment for closer attention.

INTERPRETATION

Analyzing multiple customer attributes together helps uncover high-risk segments that may not be apparent when examining each factor individually. These customers could benefit from targeted retention efforts, such as proactive engagement, payment-method guidance, and incentives to consider longer-term contracts.

Key Insights Behind Customer Churn



45.84%

Contract Type

CONTRACT TYPE

Month-to-month customers have the highest churn rate at 45.84%, compared with customers on longer-term contracts.



1.74

Satisfaction

CUSTOMER SATISFACTION

Churned customers report significantly lower average satisfaction than retained customers (1.74 vs. 3.77)



47.68%

Churn Rate

CUSTOMER TENURE

Customers in their first year show the highest churn rate (47.68%), which declines as tenure increases.



58.61%

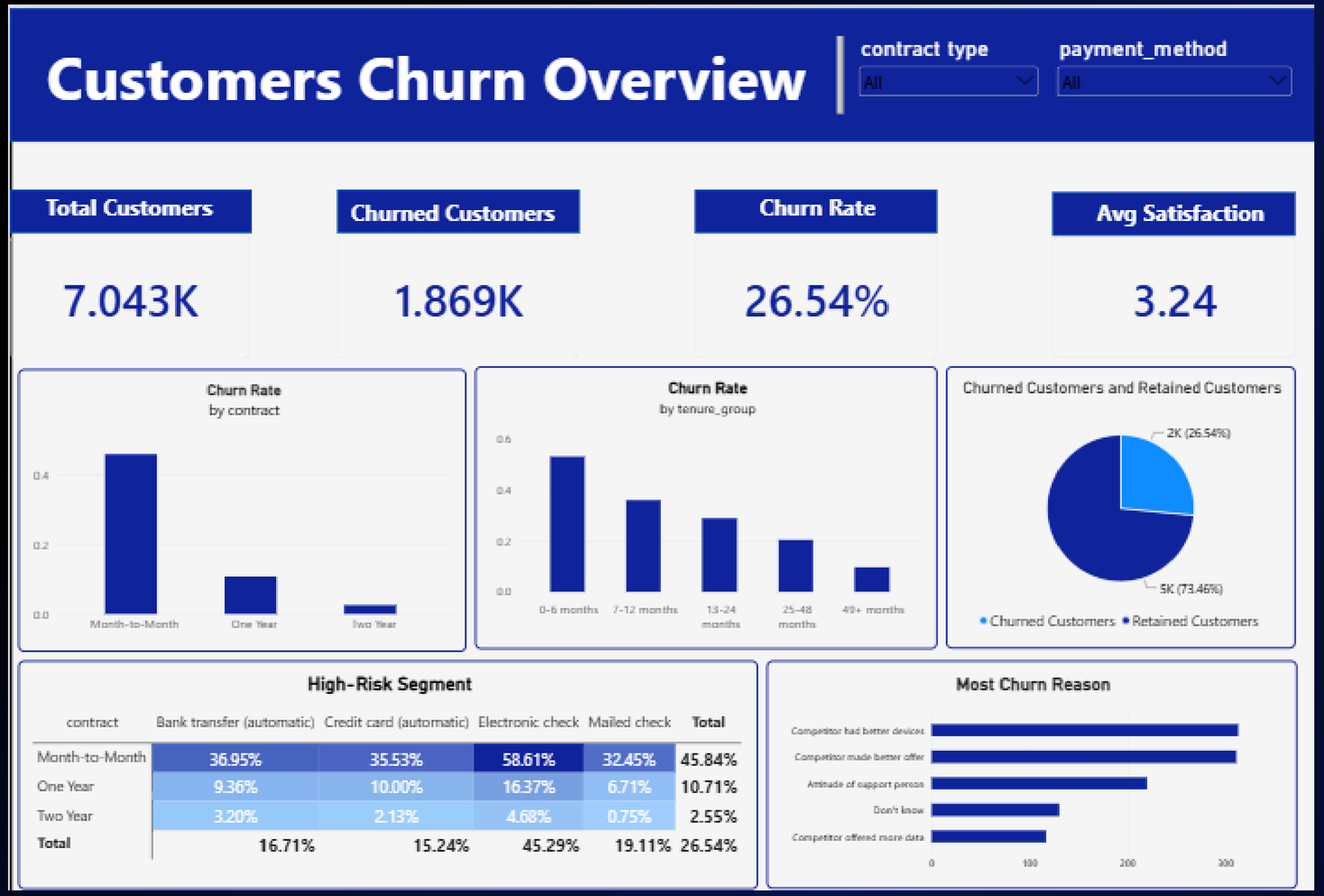
Churn Rate

HIGH-RISK SEGMENT

Customers using Electronic Check with month-to-month contracts have the highest observed churn rate at 58.61%.

Prioritize retention by proactively engaging customers with the highest risk of churn.

CUSTOMER CHURN ANALYSIS DASHBOARD



TURNING INSIGHTS INTO RETENTION STRATEGIES

Retain Month-to-Month Customers



45.84%

Contract Type

Action:

- Encourage longer-term commitments through attractive contract benefits.
- Offer loyalty incentives that reward continued subscriptions.

Strengthen Early Customer Retention



47.68%

Churn Rate

Action:

- Introduce structured onboarding programs for new customers.
- Provide proactive follow-ups and usage guidance during the first six months.

Improve Customer Satisfaction



1.74

Satisfaction

Action:

- Regularly monitor customer feedback and satisfaction levels.
- Provide proactive support to dissatisfied customers and address recurring service issues.

Prioritize High-Risk Customer Segment



58.61%

Churn Rate

Action:

- Prioritize this segment for targeted retention campaigns.
- Provide personalized support and relevant contract or payment options to address potential barriers to retention.



THANK YOU

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